

Credit Risk & Expected Loss Analytics

PD, LGD, EAD, expected loss, migration and stress testing
for a 5,000-loan retail and SME portfolio

\$8.24M

Portfolio expected loss

1.02%

EL / EAD

5,000

Loans analysed

\$17.8M

Severe-stress EL

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Model

Excel

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How to read this report. Every figure comes from the 5,000-loan synthetic data tape and the formula-driven Excel model . Historical-style outputs (default flags, recoveries, exposures) are simulated, and LGD, EAD and migration inputs are set assumptions. The value of the work is the framework and its controls, not the specific loss numbers.

1. Executive Summary

\$752.3M Total exposure	\$806.6M Total EAD	\$8.24M Portfolio expected loss	1.02% Expected loss / EAD
5,000 Loans	1.88% Observed default rate (94 defaults)	1.91% EAD-weighted average PD	45.7% EAD-weighted average LGD

This report evaluates a 5,000-loan synthetic portfolio across four products (Auto Loan, Credit Card, Retail Term Loan, SME Loan). Expected loss is built bottom-up as **PD x LGD x EAD** for every loan, using a logistic-regression PD model, product-level recovery assumptions and exposure-at-default uplifts, then aggregated by grade, product and portfolio and stressed under two adverse scenarios.

Key findings

- **Expected loss is \$8.24M, or 1.02% of EAD.** The figure reconciles to the sum of loan-level PD x LGD x EAD to within a floating-point residual (under \$0.000001), and all seven automated model controls pass.
- **Loss is concentrated in unsecured products.** Credit Card and Retail Term Loan are 59.7% of exposure but 91.6% of expected loss; the secured Auto and SME books are 40.3% of exposure and only 8.4% of loss.
- **Risk grades rank-order loss cleanly.** EL / EAD rises from 0.37% (AA) to 2.56% (E), about 7x, and observed default rates rise monotonically. Grades D and E hold 21.3% of exposure but 42.4% of expected loss.
- **Delinquency is the sharpest warning signal.** Loans with any past due (buckets 1-5) are 27.1% of exposure and 51.2% of expected loss; EL / EAD climbs from 0.68% in bucket 0 to 4.00% in bucket 5.
- **Stress raises loss materially.** Moderate stress lifts expected loss to \$11.76M (+43%); severe stress to \$17.80M (+116%), an incremental \$9.57M.
- **The PD model works but is modest.** Hold-out ROC-AUC is 0.669 (Gini 0.34). The riskiest decile captures 32% of defaults, and portfolio-level calibration is close (average PD 1.90% vs. 1.88% observed), but product-level calibration is uneven (Section 4).

Bottom line. The framework is internally consistent, reconciled and governed. The main improvements it points to are product-aware PD calibration, loan-level stress testing and calibrating LGD / EAD / migration to observed data (Section 10).

2. Scope, Data and Approach

The analysis follows the workflow of a bank credit-risk team, from a loan tape to a management view of portfolio loss.

Loan portfolio	Risk segmentation	PD model	LGD and EAD	Expected loss	Migration	Stress testing	Dashboard and audit
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Data tape

The tape has 5,000 loans originated between 2022 and 2026. It is **entirely synthetic**: no real borrowers, institutions or proprietary data are used.

Group	Fields
Segments	Loan_ID, Product (4), Region (5), Industry (6), Vintage_Year (2022-2026)
Borrower / loan features	Annual_Income, Credit_Score, DTI, Tenure_Months, Collateral_Coverage, Exposure, Delinquency_Bucket (0-5)
Risk outputs	Risk_Grade (AA, A, B, C, D, E), Default_Flag, PD_Model, LGD, EAD, Expected_Loss

Definitions

- **PD**: probability of default within 12 months, from a logistic regression on Credit Score, DTI, Delinquency Bucket and Collateral Coverage.
- **LGD**: share of exposure lost after recoveries; product-level recovery assumptions adjusted at loan level for collateral coverage.
- **EAD**: exposure expected to be outstanding at default; loan-level uplift on drawn exposure.
- **EL** = PD x LGD x EAD, computed per loan and summed. **Loss rate** = EL / EAD.

Conventions

Currency is shown in dollars. PD and default-rate comparisons use simple loan averages unless labelled EAD-weighted. Stress loss rates are stressed EL divided by base EAD. Figures are rounded for display; the workbook holds full precision.

3. Portfolio Profile and Segmentation

Exposure is spread across all four products, with Retail Term Loans the largest book. Expected loss is far less evenly distributed than exposure.

Product	Loans	Exposure	Exp. %	Default rate	Avg PD	Avg LGD	Expected loss	EL / EAD
Retail Term Loan	1,804	\$266.6M	35.4%	1.94%	2.04%	68.0%	\$4.09M	1.48%
Credit Card	1,206	\$182.6M	24.3%	1.99%	2.18%	67.7%	\$3.45M	1.61%
Auto Loan	1,245	\$192.7M	25.6%	1.20%	1.60%	10.6%	\$0.42M	0.21%
SME Loan	745	\$110.4M	14.7%	2.68%	1.58%	11.7%	\$0.27M	0.24%
Portfolio	5,000	\$752.3M	100.0%	1.88%	1.90%	45.7%	\$8.24M	1.02%

Avg PD is the simple loan average; Avg LGD is EAD-weighted.

Exposure share vs. expected-loss share by product

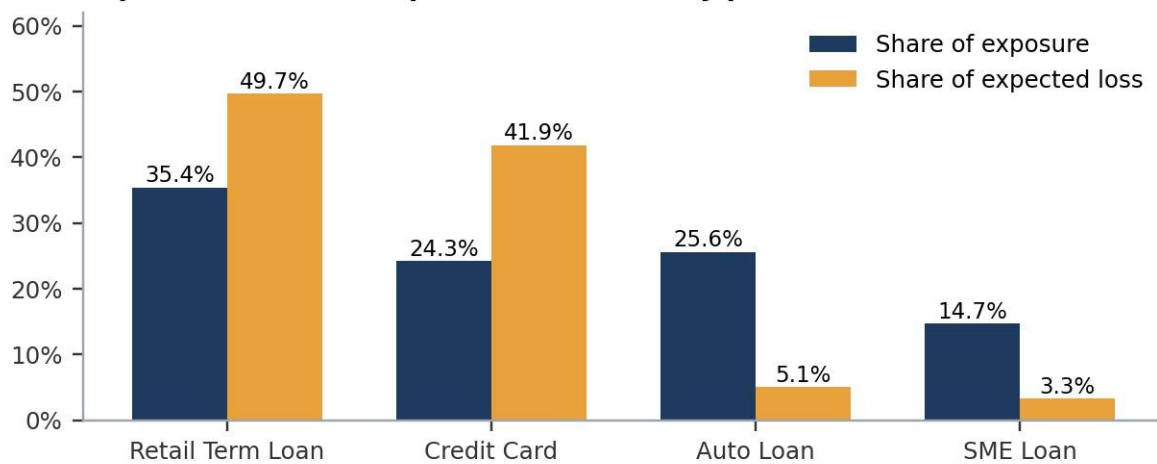


Figure 1. Unsecured products carry a disproportionate share of expected loss.

Risk-grade profile

Grade	Loans	Exposure	Exp. %	Default rate	Avg PD	Avg LGD	Expected loss	EL / EAD
AA	794	\$114.7M	15.2%	0.50%	0.67%	46.2%	\$0.46M	0.37%
A	902	\$136.1M	18.1%	1.00%	1.01%	43.7%	\$0.76M	0.52%
B	1,192	\$184.4M	24.5%	1.76%	1.49%	46.7%	\$1.64M	0.83%
C	1,057	\$157.1M	20.9%	1.89%	2.14%	45.2%	\$1.89M	1.12%
D	853	\$130.5M	17.3%	3.40%	3.35%	47.3%	\$2.68M	1.90%
E	202	\$29.5M	3.9%	5.45%	5.61%	42.5%	\$0.81M	2.56%
Total	5,000	\$752.3M	100.0%	1.88%	1.90%	45.7%	\$8.24M	1.02%

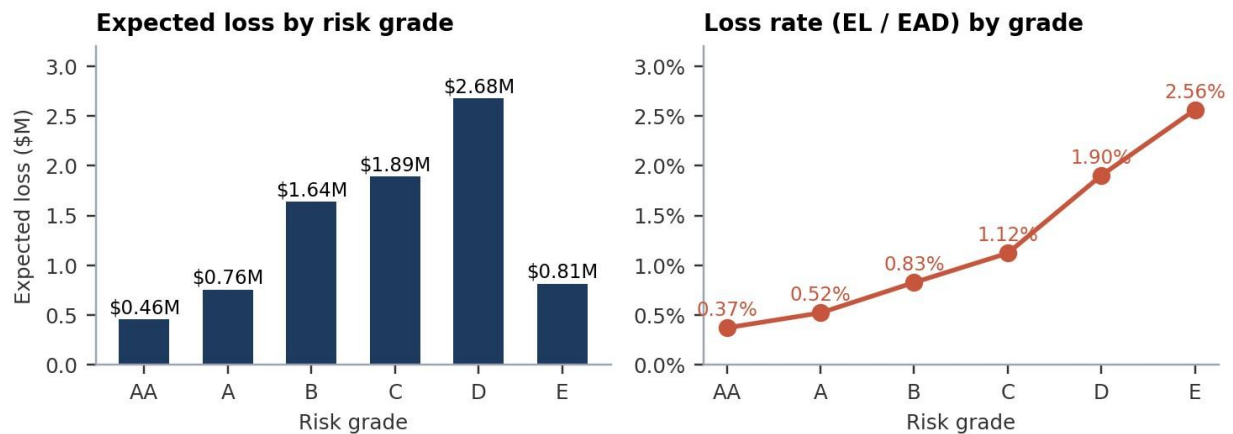


Figure 2. Loss rate rises steadily from grade AA to E; grade D contributes the most absolute loss.

Delinquency and vintage

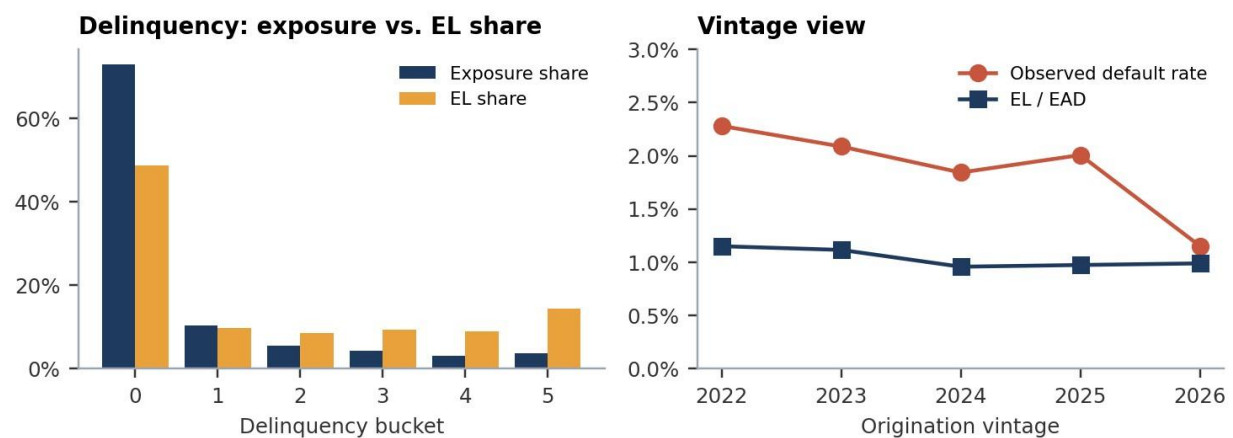


Figure 3. Past-due loans carry outsized loss; vintage default rates fall for 2026, which reflects limited seasoning rather than proven improvement.

Delinquency bucket	Loans	Exposure %	Default rate	Avg PD	EL share	EL / EAD
Bucket 0	3,656	72.9%	1.61%	1.54%	48.8%	0.68%
Bucket 1	515	10.4%	2.52%	1.86%	9.8%	0.96%
Bucket 2	282	5.5%	1.42%	2.49%	8.6%	1.58%
Bucket 3	206	4.3%	2.43%	3.10%	9.4%	2.20%
Bucket 4	147	3.1%	3.40%	3.92%	8.9%	2.89%
Bucket 5	194	3.7%	4.12%	5.00%	14.5%	4.00%

Region and industry concentration

Segment	Exposure %	EL share	EL / EAD
Region: Central	19.5%	19.9%	1.05%
Region: East	19.8%	18.8%	0.97%
Region: North	20.6%	20.5%	1.01%
Region: South	20.0%	21.1%	1.07%
Region: West	20.1%	19.6%	1.00%
Industry: Healthcare	16.7%	18.2%	1.11%
Industry: Manufacturing	16.8%	15.5%	0.94%
Industry: Other	16.1%	16.4%	1.04%
Industry: Retail	16.9%	17.7%	1.07%
Industry: Services	17.2%	16.1%	0.96%
Industry: Technology	16.2%	16.1%	1.01%

Concentration reading. Regions each hold 19.5%-20.6% of exposure and industries 16.1%-17.2%, with near-identical loss rates, so there is **no meaningful geographic or sector concentration**. This is expected for synthetic data with uniformly drawn attributes and should not be read as a real-book conclusion. Single-name risk is also low: the ten largest loans are 1.3% of exposure (largest: \$1.50M). Loss is more concentrated than balance: the top 5% of loans by EL generate 39% of expected loss.

4. Probability of Default Model

A binary logistic regression predicts default within 12 months, estimated by maximum likelihood on a stratified 70:30 train / test split. The four drivers are standard credit-quality indicators.

Driver	Coefficient	Odds ratio (per unit)	Interpretation
Credit_Score	-0.009688	0.990	Each 50-point higher score cuts default odds by about 38%
DTI	+0.032063	1.033	Each 10-point higher DTI raises default odds by about 38%
Delinquency_Bucket	+0.263326	1.301	Each extra past-due bucket raises default odds by about 30%
Collateral_Coverage	-0.145200	0.865	Higher collateral coverage lowers default odds

Signs are all economically sensible. The tape carries 94 defaults, so coefficients are estimated on a small event count and should be read as illustrative.

Discrimination and accuracy

Metric	Value	Comment
ROC-AUC (hold-out)	0.669	Modest discrimination, typical of a four-variable illustrative model
Gini (2 x AUC - 1)	0.337	Derived from hold-out AUC
Brier score	0.0183	Barely better than a constant-rate baseline of 0.0184; expected at a 1.9% default rate
Full-tape AUC (in-sample)	0.710	Optimistic by construction; shown to make the train / test gap visible
Average PD vs. observed	1.90% vs. 1.88%	Portfolio-level calibration is close

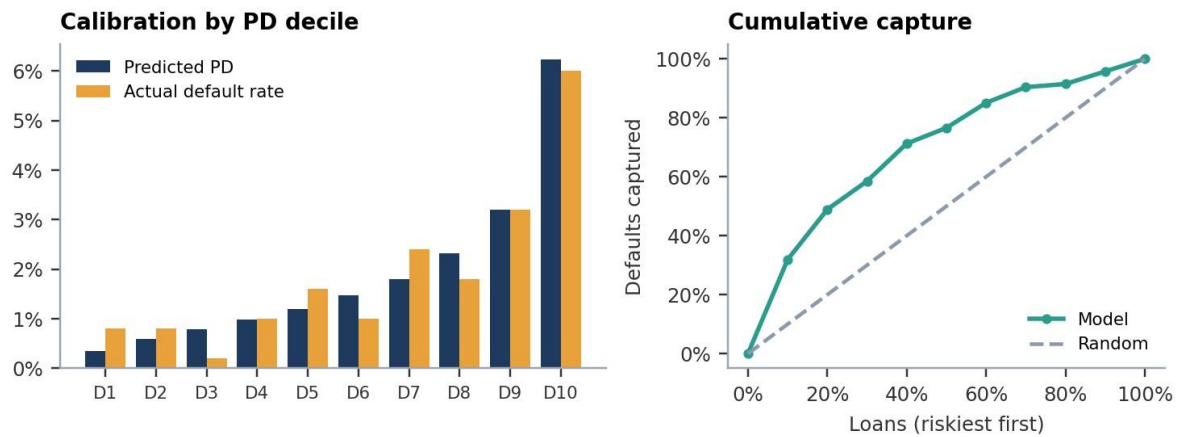


Figure 4. Decile calibration and cumulative capture. The riskiest 10% of loans capture 31.9% of defaults, a 3.2x lift.

Decile	Loans	Avg predicted PD	Actual defaults	Actual rate	Actual / predicted
D1	500	0.35%	4	0.80%	2.27
D2	500	0.59%	4	0.80%	1.35
D3	500	0.79%	1	0.20%	0.25
D4	500	0.99%	5	1.00%	1.01
D5	500	1.20%	8	1.60%	1.33
D6	500	1.47%	5	1.00%	0.68
D7	500	1.80%	12	2.40%	1.33
D8	500	2.33%	9	1.80%	0.77
D9	500	3.20%	16	3.20%	1.00
D10	500	6.23%	30	6.00%	0.96

Deciles are ordered from lowest (D1) to highest (D10) predicted PD. With 0-30 defaults per decile, ratios are noisy: D3 (1 default) and D7 (12 defaults) deviate mainly from sampling variation.

Calibration by product

The model has no product term, so calibration can drift across products. SME is the clearest gap.

Product	Loans	Defaults	Observed rate	Average PD	Observed / predicted
Auto Loan	1,245	15	1.20%	1.60%	0.75
Credit Card	1,206	24	1.99%	2.18%	0.91
Retail Term Loan	1,804	35	1.94%	2.04%	0.95
SME Loan	745	20	2.68%	1.58%	1.70

Finding. SME loans default at 2.68% against an average model PD of 1.58% (20 defaults vs. about 12 expected; roughly a 2% one-sided Poisson tail probability). Auto loans are over-predicted (1.20% observed vs. 1.60%). Because SME has low LGD, the dollar effect is small today, but a product-aware PD or a product calibration factor would remove the bias.

Rank-ordering by risk grade

Observed default rates rise monotonically across grades: AA 0.50%, A 1.00%, B 1.76%, C 1.89%, D 3.40%, E 5.45%. Grade-level average PDs track observed rates closely, apart from small-sample differences in grades AA and B.

5. Loss Given Default and Exposure at Default

LGD is defined as **1 - recovery rate**. Secured facilities (Auto, SME) recover about 89% of exposure; unsecured facilities (Credit Card, Retail Term) recover about 32%. At loan level, LGD falls as collateral coverage rises (correlation between -0.41 and -0.58 within each product), bounded between 5% and 98%.

Product	Facility type	Collateral	Recovery	Standard LGD	Mean loan LGD	EAD / exposure
Auto Loan	Secured retail	Vehicle title	89.3%	10.7%	10.7%	1.040x
Credit Card	Unsecured revolving	None	32.2%	67.8%	67.8%	1.177x
Retail Term Loan	Unsecured amortizing	None	32.3%	67.7%	67.7%	1.039x
SME Loan	Secured commercial	Equipment and real estate	88.8%	11.2%	11.2%	1.040x

Exposure at default lifts drawn exposure by about 4% for term and secured products and about 18% for credit cards, reflecting drawdown behaviour on revolving lines ahead of default. Loan-level ratios range from 1.00x to 1.08x (term products) and 1.05x to 1.30x (cards).

Documentation note. The tape has no undrawn-limit field, so EAD is an assumption-based uplift on exposure and cannot be re-derived from the conversion factors in the assumptions table (Auto 20%, Card 50%, Retail Term 0%, SME 30%). Those factors document intent; the near-identical 1.04x ratios for Auto, Retail Term and SME show they do not drive EAD mechanically.

6. Expected Loss

Expected loss is computed per loan as $PD \times LGD \times EAD$ from the stored, rounded parameters, then aggregated. This is the V2 correction: the first build computed EL from higher-precision intermediates than the values it stored. The loan-level tie-out now agrees to within a floating-point residual of 6×10^{-11} dollars.

Level	Expected loss	EL / EAD	Comment
Portfolio	\$8.236M	1.02%	5,000 loans, \$806.6M EAD
Highest-loss grade	D: \$2.68M	1.90%	32.5% of portfolio EL
Highest loss-rate grade	E: \$0.81M	2.56%	Smallest grade by exposure (3.9%)
Highest-loss product	Retail Term: \$4.09M	1.48%	49.7% of portfolio EL
Highest loss-rate product	Credit Card: \$3.45M	1.61%	Highest EAD uplift and LGD

What drives the loss

- **Structure beats size.** Secured products carry an LGD of roughly 11% against roughly 68% for unsecured ones, which outweighs their similar or higher default rates.
- **Risk is back-loaded in the tails.** The top 1% of loans (50 loans) generate 16% of EL and the top 5% generate 39%, so a few high-PD, high-LGD, high-EAD loans matter.
- **Early warning is measurable.** Past-due loans (buckets 1-5) are 27.1% of exposure but 51.2% of EL, so cure and roll-rate management directly lowers expected loss.

Migration

The 1-year transition matrix below is an **assumed, illustrative** matrix, not estimated from observed cohorts. Each row sums to 100%. Ratings are sticky (77-88% stay in grade), downgrades are more likely than upgrades in every grade, and default risk rises sharply toward the weakest grades (0.2% for AA, 12.0% for E).

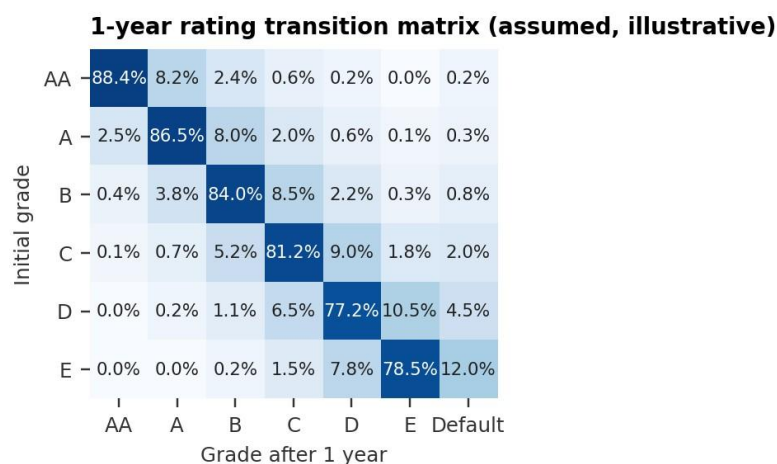


Figure 5. Transition probabilities by starting grade, including default.

Starting grade	P(upgrade)	P(stay)	P(downgrade)	P(default)	Net drift
AA	0.0%	88.4%	11.4%	0.2%	-11.6 pp
A	2.5%	86.5%	10.7%	0.3%	-8.5 pp
B	4.2%	84.0%	11.0%	0.8%	-7.6 pp
C	6.0%	81.2%	10.8%	2.0%	-6.8 pp
D	7.8%	77.2%	10.5%	4.5%	-7.2 pp
E	9.5%	78.5%	0.0%	12.0%	-2.5 pp

Net drift = $P(\text{upgrade}) - P(\text{downgrade}) - P(\text{default})$. Grade AA has no upgrade path, so its drift is negative by construction.

7. Stress Testing

Two adverse scenarios apply shocks to PD (multiplier), LGD (absolute add-on) and EAD (multiplier). The Base case reconciles exactly to portfolio expected loss.

Scenario	PD mult.	LGD add-on	EAD mult.	Stressed EL	Incremental	Uplift	EL / base EAD
Base	1.00x	+0 pp	1.00x	\$8.24M	-	-	1.02%
Moderate Stress	1.25x	+5 pp	1.03x	\$11.76M	\$3.53M	+42.8%	1.46%
Severe Stress	1.60x	+12 pp	1.07x	\$17.80M	\$9.57M	+116.2%	2.21%

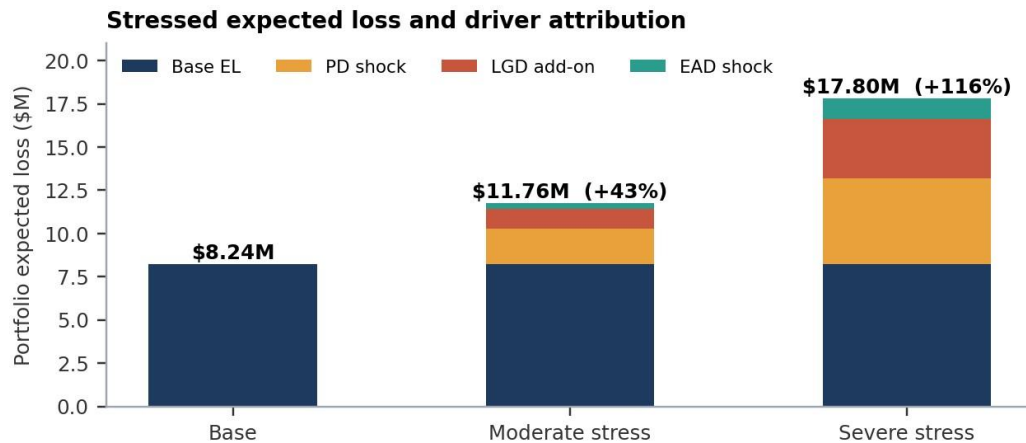


Figure 6. Driver attribution is sequential (PD, then LGD, then EAD) and therefore order-dependent.

In the moderate scenario PD accounts for \$2.06M of the \$3.53M increase, LGD \$1.13M and EAD \$0.34M. In the severe scenario the split is \$4.94M, \$3.46M and \$1.16M of \$9.57M. PD is the largest lever in both cases.

Sensitivity grid: PD shock x LGD add-on

PD shock \ LGD add-on	+0 pp	+3 pp	+5 pp	+8 pp	+10 pp	+12 pp
1.00x	\$8.24M	\$8.78M	\$9.14M	\$9.68M	\$10.04M	\$10.40M
1.15x	\$9.47M	\$10.09M	\$10.51M	\$11.13M	\$11.54M	\$11.96M
1.25x	\$10.29M	\$10.97M	\$11.42M	\$12.10M	\$12.55M	\$13.00M
1.40x	\$11.53M	\$12.29M	\$12.79M	\$13.55M	\$14.05M	\$14.56M
1.60x	\$13.18M	\$14.04M	\$14.62M	\$15.48M	\$16.06M	\$16.64M

Grid shows portfolio EL before any EAD shock, using the workbook's aggregate method.

Segment impact (loan-level cross-check)

Re-scoring every loan under the same shocks (PD scaled, LGD add-on capped at 100%, EAD scaled) lets the impact be split by product and cross-checks the workbook's portfolio-level shortcut.

Product	Base EL	Moderate EL	Uplift	Severe EL	Uplift
Retail Term Loan	\$4.09M	\$5.59M	+37%	\$7.96M	+95%
Credit Card	\$3.45M	\$4.71M	+37%	\$6.68M	+94%
Auto Loan	\$0.42M	\$0.75M	+78%	\$1.38M	+228%
SME Loan	\$0.27M	\$0.47M	+71%	\$0.84M	+206%
Portfolio (loan-level)	\$8.24M	\$11.52M	+40%	\$16.86M	+105%

Two takeaways. (1) The workbook's aggregate method is mildly conservative: it gives \$11.76M and \$17.80M versus \$11.52M and \$16.86M at loan level (+2.2% and +5.6%), because it scales by EAD-weighted rather than PD-weighted LGD. (2) A flat percentage-point LGD add-on hits secured products hardest in relative terms (Auto and SME loss rise by roughly 3x under severe stress) because their base LGD is low, yet unsecured products still dominate the dollar increase. A collateral-value shock would be a more realistic way to stress secured LGD.

8. Model Governance and Controls

Seven automated controls run in the workbook (sheet 08). Their criteria were re-evaluated independently against the data tape while building this report.

ID	Control	Criterion	Severity	Result
CTL-01	PD bounds	$0 \leq \text{PD} \leq 1$ across all loans	High	PASS
CTL-02	LGD bounds	$0 \leq \text{LGD} \leq 1$ across all loans	High	PASS
CTL-03	Positive exposure & EAD	Exposure > 0 and EAD > 0	Critical	PASS
CTL-04	Expected-loss tie-out	Max PD x LGD x EAD - loan EL (< \$1 in total)	Critical	PASS
CTL-05	Stress base tie-out	Base stressed EL = portfolio EL	Critical	PASS
CTL-06	Data completeness	5,000 records, no nulls	Medium	PASS
CTL-07	Migration row sums	Each transition row sums to 100%	High	PASS

Key reconciliation: sum of loan-level PD x LGD x EAD = \$8.235783M, matching the reported portfolio EL; the maximum loan-level difference is 6.2e-11 dollars.

Version history: findings corrected in V2

#	V1 issue	V2 correction
1	Expected loss was computed from higher-precision intermediates than the rounded PD / LGD / EAD values it stored, causing a tie-out mismatch.	EL is recalculated from the stored values and a loan-level tie-out control (CTL-04) was added.
2	PD_Model was the latent probability used to generate the synthetic default flag, so the PD 'model' was not a real fit.	An illustrative logistic regression on Credit Score, DTI, Delinquency Bucket and Collateral Coverage is fitted; its predictions drive PD and EL.
3	Stress scenarios were not tied to the corrected loss figure.	Base scenario reconciles to portfolio EL (CTL-05); Moderate and Severe apply transparent PD, LGD and EAD shocks.

9. Limitations

- **Synthetic data.** Defaults, recoveries and attribute relationships are simulated. Region / industry flatness and default patterns do not describe any real book.
- **Modest PD discrimination and a low event count.** AUC 0.669 and 94 defaults limit precision; decile calibration is noisy and out-of-time validation was not performed.
- **No product term in PD.** Product-level calibration is uneven (SME under-predicted, Auto over-predicted).
- **Assumption-based LGD, EAD and migration.** Recoveries and EAD uplifts are set, not calibrated to workout data; conversion factors cannot be re-derived from the tape; the migration matrix is illustrative.
- **Aggregate stress in Excel.** The workbook scales portfolio EL and does not project loans under macro paths; a loan-level cross-check is provided here but is not macro-linked.
- **Not regulatory-compliant.** The framework borrows IFRS 9 / Basel and DFAST / CCAR vocabulary (12-month PD, PD x LGD x EAD, downturn LGD, adverse scenarios) but has no staging, lifetime ECL, forward-looking overlays or capital calculation.

10. Suggested Next Steps

Priority	Action	Benefit
1	Add product (or a product calibration factor) to the PD model; validate on an out-of-time sample	Removes SME / Auto bias; more defensible discrimination metrics
2	Move stress testing to loan level in Python with a collateral-value shock for secured LGD	Segment-accurate stress and more realistic secured-portfolio behaviour
3	Add an undrawn-limit field and derive EAD from CCFs	Makes EAD traceable and auditable
4	Estimate migration from observed cohorts and add roll-rate analysis by delinquency bucket	Evidence-based transitions; cure and roll dynamics
5	Add IFRS 9 staging and lifetime ECL as an extension	Closer alignment with provisioning practice

Appendix: Methodology Reference

Formulas

Item	Definition
Expected loss	$EL(i) = PD(i) \times LGD(i) \times EAD(i)$; portfolio EL = sum of loan EL
Loss rate	EL / EAD
PD model	$\text{logit}(PD) = \text{intercept} + b1 \times \text{Credit_Score} + b2 \times DTI + b3 \times \text{Delinquency_Bucket} + b4 \times \text{Collateral_Coverage}$
Gini	$2 \times AUC - 1$
Brier score	Mean of $(PD - \text{default flag})^2$; constant-rate baseline = $p \times (1 - p)$
LGD	1 - recovery rate; loan-level LGD adjusted for collateral coverage, bounded 5%-98%
EAD	Exposure x uplift (about 1.04x term and secured products, about 1.18x credit cards)
Stressed EL (workbook)	Base EL x PD mult. x (weighted LGD + add-on) / weighted LGD x EAD mult.
Stressed EL (loan level)	Sum of $\min(PD \times m, 1) \times \min(LGD + a, 1) \times EAD \times e$

Deliverables

Artifact	Purpose
Model/Credit_Risk_Expected_Loss_Model_Professional.xlsx	Nine-sheet formula-driven workbook (summary, analytics, PD, LGD / EAD, EL, migration, stress, controls, data tape)
Data/Credit_Risk_Portfolio_Analytics.csv	5,000-loan synthetic data tape
Python/credit_risk_pd_model.py	PD model, validation, EL calculation, stress testing
SQL/credit_risk_queries.sql	Default-rate, grade, delinquency, concentration, vintage and EL queries
Methodology/	methodology.md

Disclaimer. This report is an educational and portfolio demonstration built on synthetic data. It contains no real borrower or institutional information and is not lending, investment, accounting or regulatory advice.